KAIZEN MATHS

Lesson starter

Complex Numbers

Powers, Roots and Equations

Name _____

S1

Use De Moivre theorem to find the power.

Use De Moivre's theorem to find $[3(\cos \pi/3 + i \sin \pi/3)]^3$.

S2

Find all roots in polar form.

Find all solutions of $z^5 = 243\ \text{cis}(\pi/2)$.

S3

Solve each quadratic equation over the complex numbers.

Solve $z^2 - 6z + 18 = 0$.

S4

Solve each equation for z.

Solve $(2 + 3i)z = 9 - 6i$.

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ALG-D7A3-L3 · S

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KAIZEN MATHS

Exit ticket

Complex Numbers

Powers, Roots and Equations

Name

E1

Use De Moivre theorem to find the power.

Use De Moivre's theorem to find $[2(\cos \pi/3 + i \sin \pi/3)]^4$.

E2

Find all roots in polar form.

Find all solutions of $z^3 = 8\ \text{cis}(\pi/2)$.

E3

Solve each quadratic equation over the complex numbers.

Solve $z^2 + 6z + 45 = 0$.

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Teacher answers & working

Complex Numbers · Powers, Roots and Equations

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S1

Use De Moivre theorem to find the power.

Use De Moivre's theorem to find $[3(\cos \pi/3 + i \sin \pi/3)]^3$.

Answer:

$27(\cos \pi + i \sin \pi)$

Working:

De Moivre's theorem: $[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$.

Here r = 3, θ = π/3, and n = 3.

$r^n = 3^3 = 27$.

$n\theta = 3 \times \pi/3 = \pi$.

Therefore the value is 27(cos π + i sin π).

S2

Find all roots in polar form.

Find all solutions of $z^5 = 243\ \text{cis}(\pi/2)$.

Answer:

3 cis(0.314 rad), 3 cis(π/2), 3 cis(2.827 rad), 3 cis(4.084 rad), 3 cis(5.341 rad)

Working:

Write z = r cis(α). Then $z^5 = r^5\ \text{cis}(5\alpha)$.

Compare moduli: $r^5 = 243$, so r = 3.

Compare arguments: 5α = π/2 + 2kπ.

So $\alpha = (\pi/2 + 2k\pi)/5$, for $k = 0, 1, ..., 4$.

Substitute each value of k into α = (π/2 + 2kπ)/5.

The roots are 3 cis(0.314 rad), 3 cis(π/2), 3 cis(2.827 rad), 3 cis(4.084 rad), 3 cis(5.341 rad).

S3

Solve each quadratic equation over the complex numbers.

Solve $z^2 - 6z + 18 = 0$.

Answer:

$z = 3 + 3i$ or $z = 3 - 3i$

Working:

Use the quadratic formula $z = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$.

Here a = 1, b = -6, and c = 18.

Discriminant = $b^2 - 4ac = (-6)^2 - 4(1)(18) = -36$.

$\sqrt{-36} = 6i$ because $-36 = -36$.

$z = \frac{6 \pm 6i}{2} = 3 + 3i$ or $3 - 3i$.

Teacher answers & working

Complex Numbers · Powers, Roots and Equations

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S4

Solve each equation for z.

Solve (2 + 3i)z = 9 - 6i.

Answer:

$z = -3i$

Working:

Divide both sides by 2 + 3i.

$z = \frac{9-6i}{2+3i}$.

Multiply numerator and denominator by the conjugate 2 - 3i.

Denominator = $2^2 + 3^2 = 13$.

This simplifies to z = -3i.

E1

Use De Moivre theorem to find the power.

Use De Moivre's theorem to find $[2(\cos \pi/3 + i \sin \pi/3)]^4$.

Answer:

$16(\cos 4\pi/3 + i \sin 4\pi/3)$

Working:

De Moivre's theorem: $[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$.

Here r = 2, θ = π/3, and n = 4.

$r^n = 2^4 = 16$.

$n\theta = 4 \times \pi/3 = 4\pi/3$.

Therefore the value is 16(cos 4π/3 + i sin 4π/3).

E2

Find all roots in polar form.

Find all solutions of $z^3 = 8\ \text{cis}(\pi/2)$.

Answer:

$2\ \text{cis}(\pi/6),\ 2\ \text{cis}(5\pi/6),\ 2\ \text{cis}(3\pi/2)$

Working:

Write z = r cis(α). Then $z^3 = r^3\ \text{cis}(3\alpha)$.

Compare moduli: $r^3 = 8$, so r = 2.

Compare arguments: $3\alpha = \pi/2 + 2k\pi$.

So $\alpha = (\pi/2 + 2k\pi)/3$, for $k = 0, 1, ..., 2$.

Substitute each value of k into α = (π/2 + 2kπ)/3.

The roots are 2 cis(π/6), 2 cis(5π/6), 2 cis(3π/2).

Teacher answers & working

Complex Numbers · Powers, Roots and Equations

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E3

Solve each quadratic equation over the complex numbers.

Solve $z^2 + 6z + 45 = 0$.

Answer:

$z = -3 + 6i$ or $z = -3 - 6i$

Working:

Use the quadratic formula $z = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$.

Here a = 1, b = 6, and c = 45.

Discriminant = $b^2 - 4ac = 6^2 - 4(1)(45) = -144$.

$\sqrt{-144} = 12i$ because $-144 = -144$.

$z = \frac{-6 \pm 12i}{2} = -3 + 6i$ or $-3 - 6i$.